



Secure Offline Password Manager with Browser Autofill

The Password Security Crisis

In today's digital landscape, password security has become a critical vulnerability that poses a threat to both individuals and organizations. The data tells a sobering story about our collective password practices.

Critical Statistics

- **70% of data breaches** are caused by weak or reused passwords
- **100+ accounts** – the average person's digital footprint is impossible to remember securely
- **Unencrypted storage** – most browser autofill and cloud managers remain vulnerable

 **We need a password manager that is both secure and convenient** – without the trade-offs that compromise either privacy or usability.



PassGuard: Military-Grade Security Meets Simplicity

PassGuard reimagines password management by combining enterprise-level security with effortless usability. Our vision is clear: making password security accessible, local, and uncompromising – where privacy and convenience exist in perfect balance.



AES-256-GCM Encryption

Military-grade encryption ensuring zero data leaks. Your passwords are protected by the same cryptographic standards used by governments and financial institutions worldwide.



Smart Autofill

One-click secure logins via browser extension. Seamlessly integrate password security into your daily workflow without friction or complexity.



Always Available

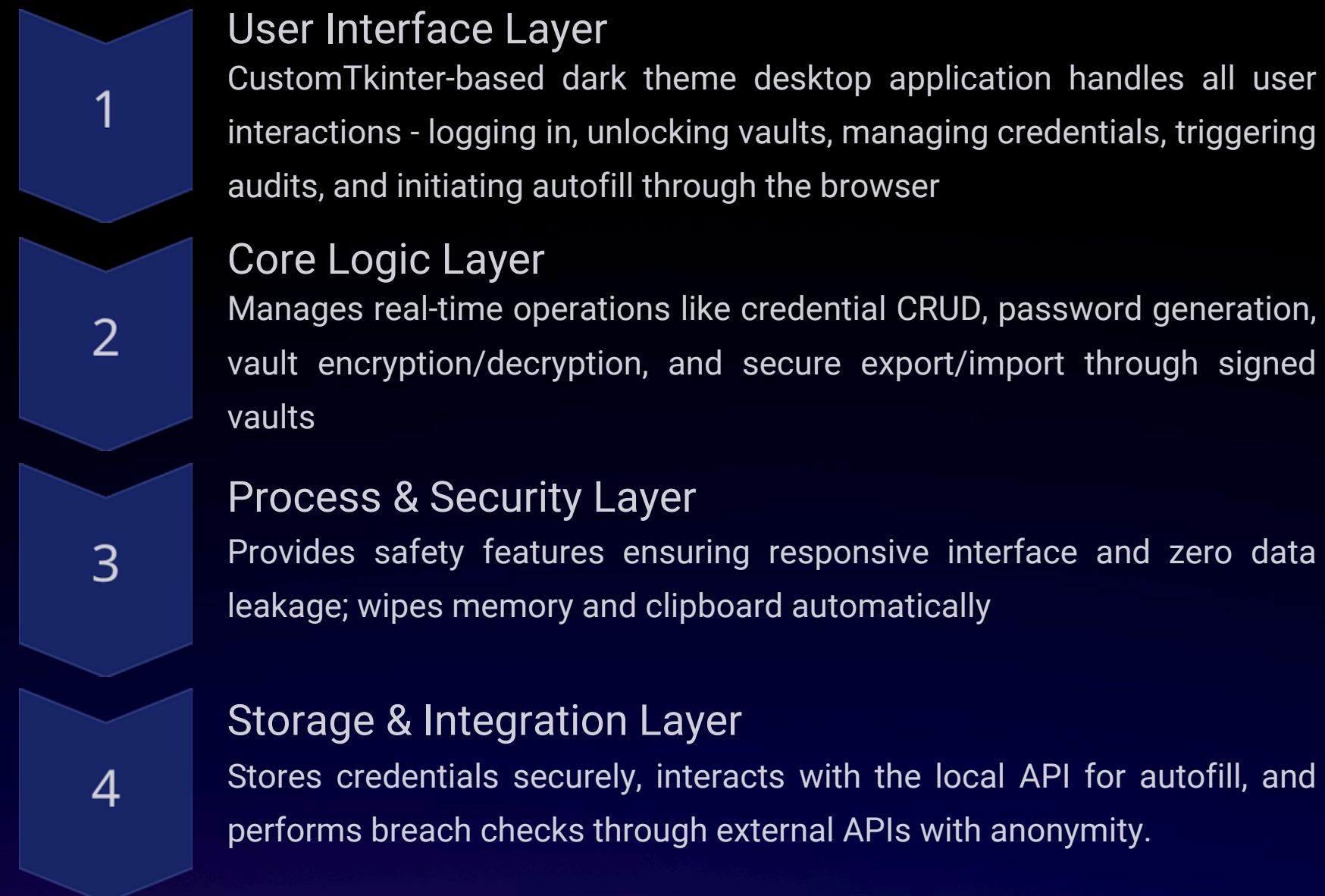
Runs from system tray for instant access without the cloud. Your data never leaves your device, ensuring complete control and privacy at all times.

"Making password security accessible, local, and uncompromising — privacy and usability in perfect balance."

Architecture + Security Foundation

PassGuard is built on a robust, layered security architecture that ensures your credentials remain protected at every level. Our technical implementation combines industry-leading encryption standards with zero-knowledge principles.

System Architecture



Tech Stack

- **Backend:** Python 3.13, Flask, Cryptography libraries
- **Frontend:** CustomTkinter (desktop), JavaScript (extension)
- **Security:** AES-256-GCM, Argon2id, RSA-4096

"Security engineered, not assumed."

Key Features: Innovation Meets Security

PassGuard delivers a comprehensive suite of features that balance cutting-edge security with practical usability. Every feature is designed to protect your digital identity while making password management effortless.



Military-Grade Encryption

AES-256-GCM with zero-knowledge design ensures only you can decrypt your data. Your master password never leaves your device, and even we cannot access your vault.



Secure Browser Autofill

Seamless browser integration using local Flask API means no credentials are ever exposed to the web. One-click login that maintains security at every step.



Security Audit System

Automatically identifies weak or reused passwords and provides comprehensive strength metrics. Integration with HaveIBeenPwned alerts you to compromised credentials.



Encrypted Credential Sharing

RSA-4096 public key exchange allows secure credential export to trusted contacts without exposing your master password or vault structure.



USB Physical Lock

Automatically locks your vault when the designated USB key is removed, providing an additional physical-layer security mechanism.

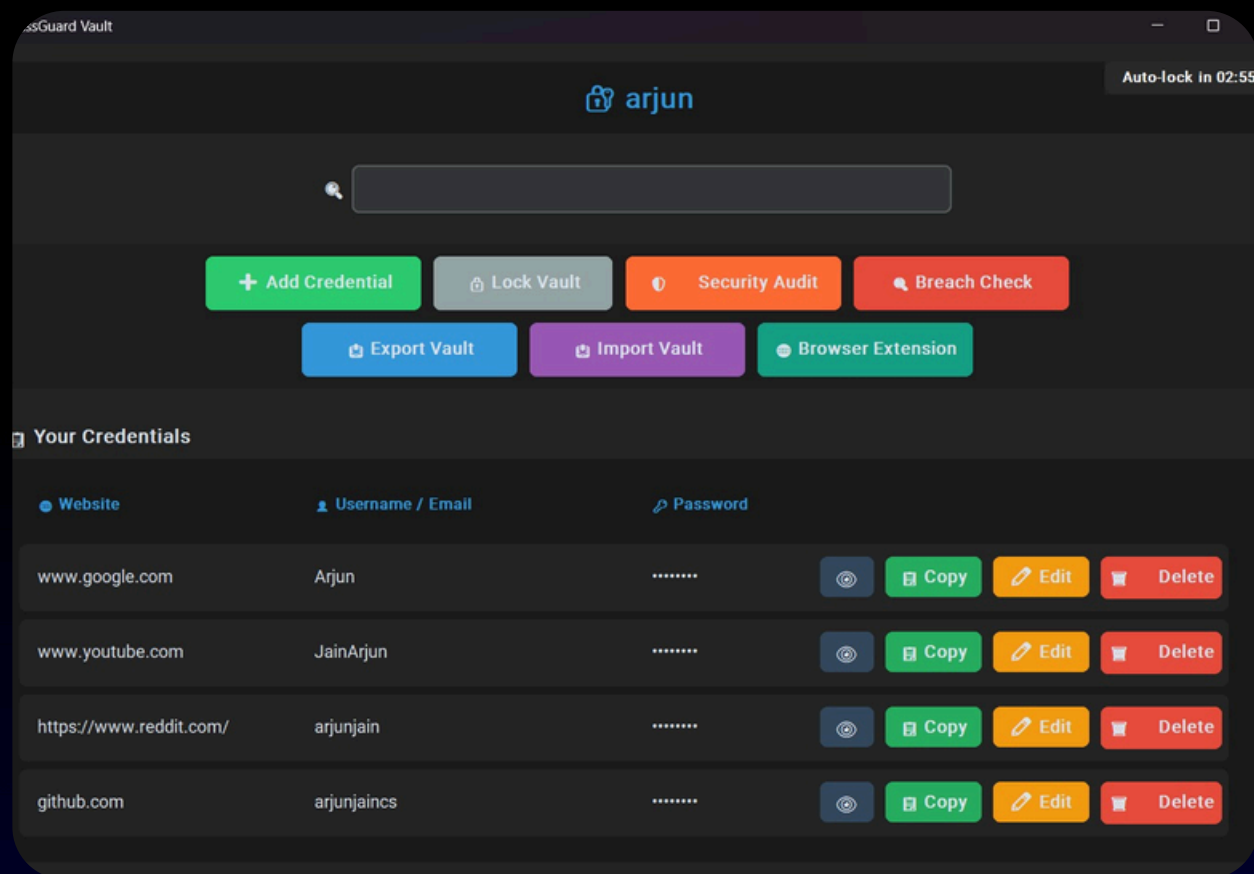


System Tray Integration

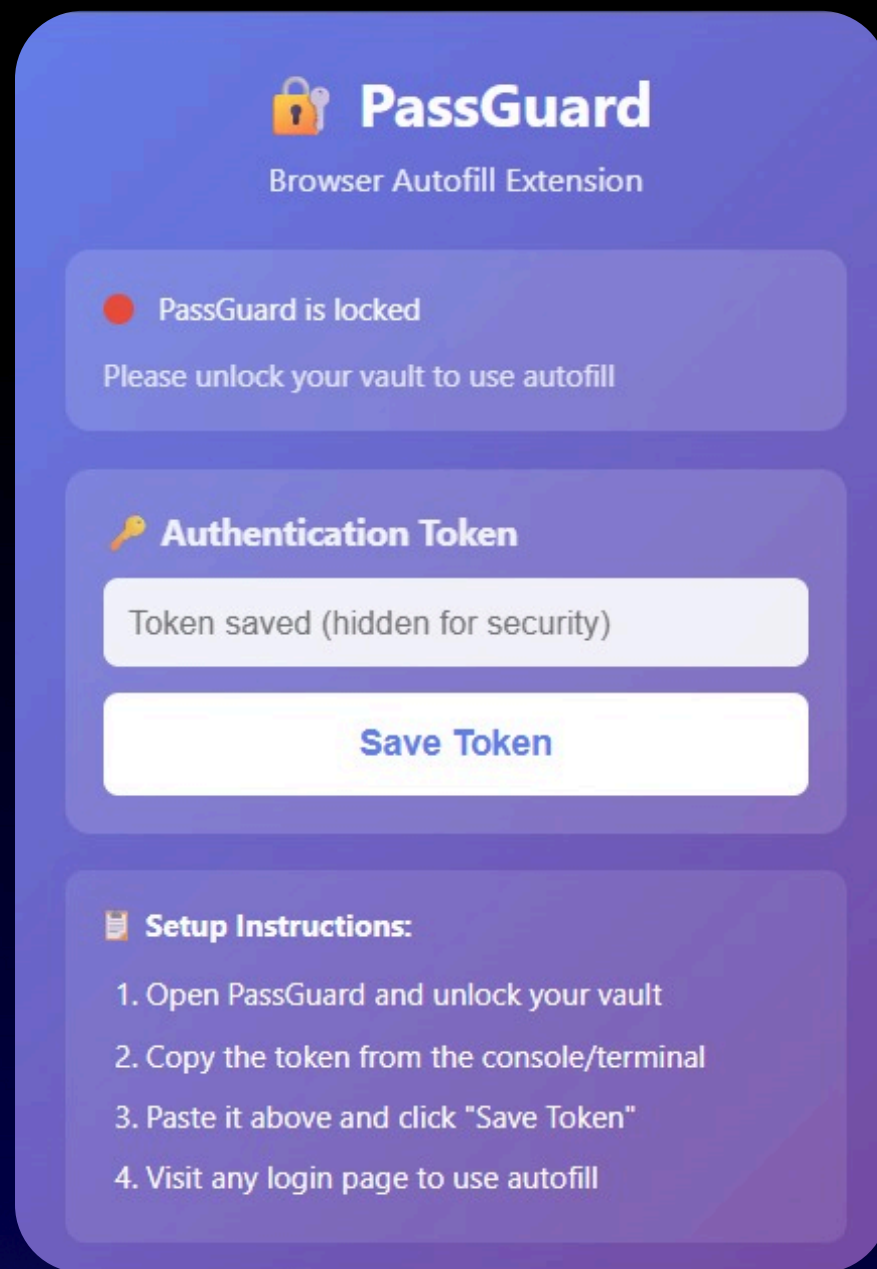
Always accessible from your system tray for quick unlock, autofill, or password generation without disrupting your workflow.

PassGuard in Action

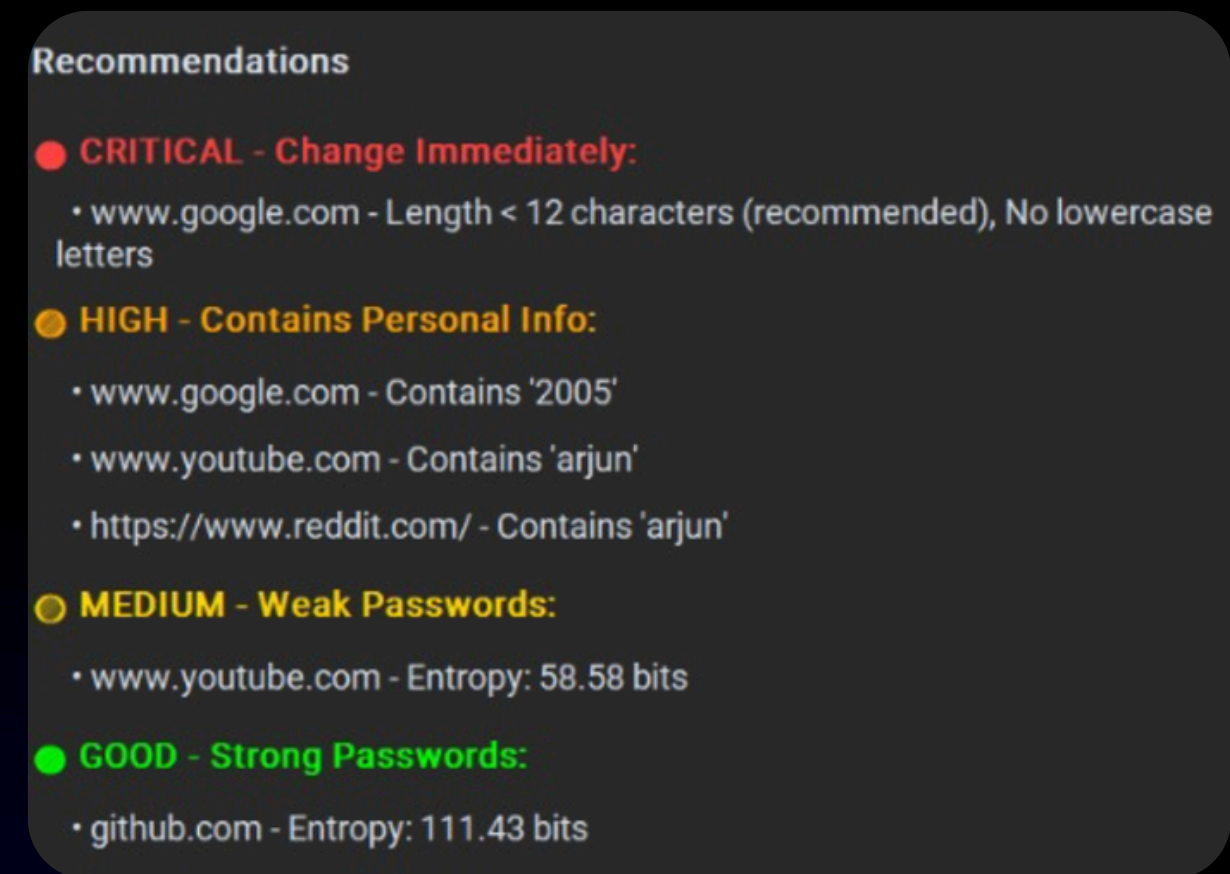
Experience the elegance and security of PassGuard through our interface design. Every element is crafted to provide maximum security with minimal friction, making password management intuitive and reliable.



Clean, intuitive dark-themed UI built with CustomTkinter allows you to manage 100+ passwords effortlessly. Organized categories, quick search, and visual password strength indicators make navigation seamless.



One-click secure login with local encryption. The Chrome extension communicates with your local vault via encrypted localhost API, ensuring credentials never touch the internet.



Real-time breach detection and strength scoring identifies vulnerabilities in your password portfolio. Get actionable insights to strengthen your security posture instantly.

Impact + Future Roadmap

PassGuard is designed to scale from individual users to enterprise teams, with a clear vision for evolution. Our roadmap ensures that PassGuard remains at the forefront of password security innovation.

Real-World Impact

Professionals

Securely manage company and client credentials with enterprise-grade encryption. Enable safe, auditable credential sharing with team members while maintaining zero-knowledge architecture.

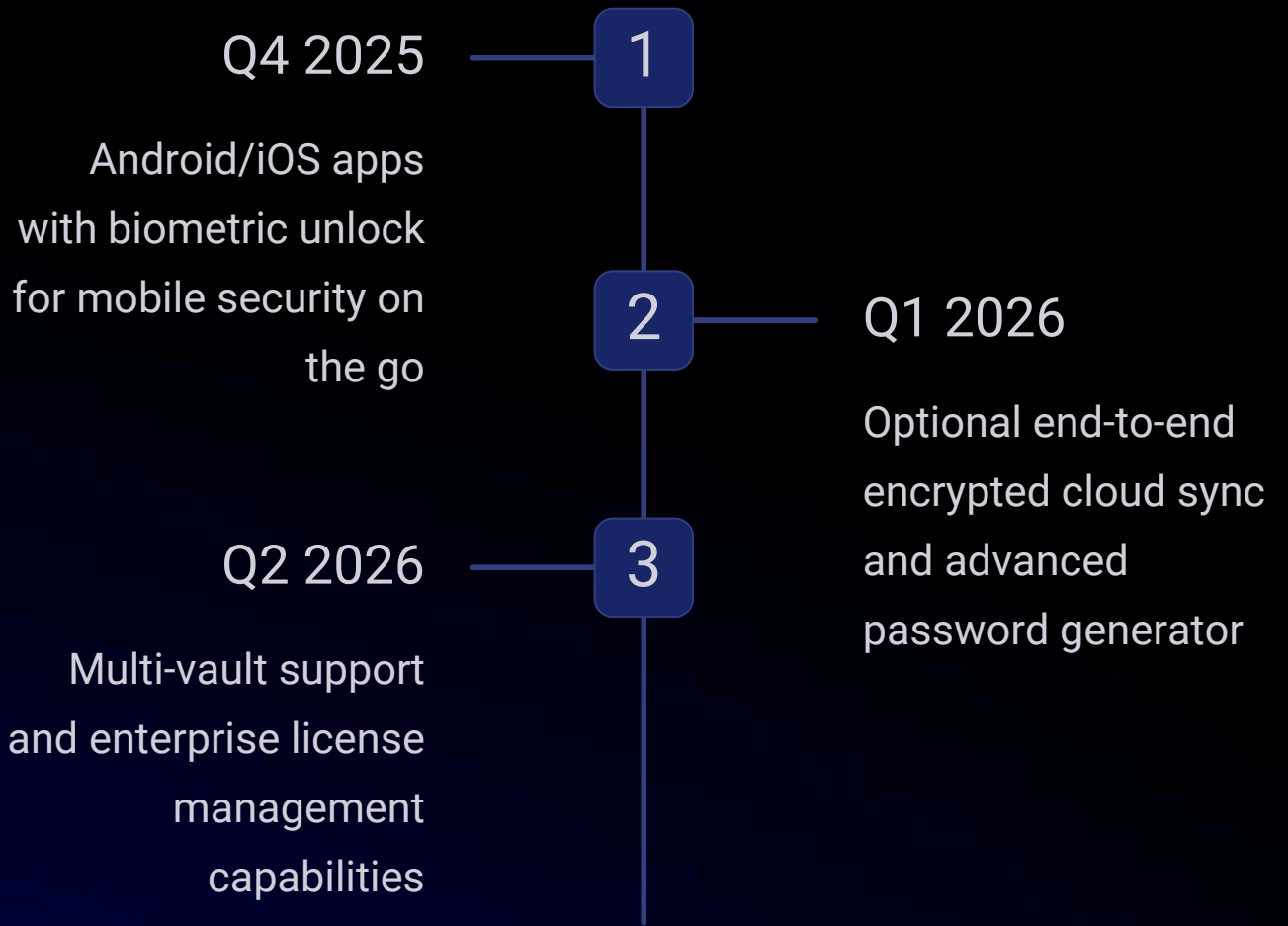
Students

Single vault for academic portals, project accounts, and collaboration tools. Never worry about remembering dozens of passwords across different platforms and services.

Families

Shared access for streaming or utility accounts with individual security boundaries. Educate family members on password hygiene while maintaining convenience and safety.

Development Timeline



PassGuard evolves — from personal privacy to team-grade password security, always maintaining our core commitment to zero-knowledge architecture.

PassGuard: Your Passwords, Truly Secure.

PassGuard drives technological innovation with a free, open-source solution built on AES-256-GCM and Argon2id encryption. Its fully offline design strengthens digital infrastructure by setting a new standard for secure, resilient data protection.

Industry, Innovation and Infrastructure (SDG 9)



By using a zero-knowledge architecture, PassGuard ensures complete privacy data never leaves the user's device. Its auditable open-source (MIT) model fosters trust, transparency, and accountability in digital security.

Peace, Justice and Strong Institutions (SDG 16)

IMPACTS

Individuals

Securely manage personal accounts offline.

Developers

Protect work credentials across platforms.

Organisations

Deploy as an internal secure vault system.

Education

Demonstrates real-world, hardware-based security.

Response Teams

Enable quick audits and breach detection.